

Assignment 2 Report

Introduction

This report focuses on creating a business intelligence pipeline to support better decision-making. The project demonstrates the process of extracting, transforming, and loading (ETL) raw data into a database, building a multidimensional cube, and developing an interactive Power BI dashboard for analysis. The dataset consists of five tables: one fact table (FactTable) and four dimension tables (Customer, Product, Store, and Time).

ETL Process

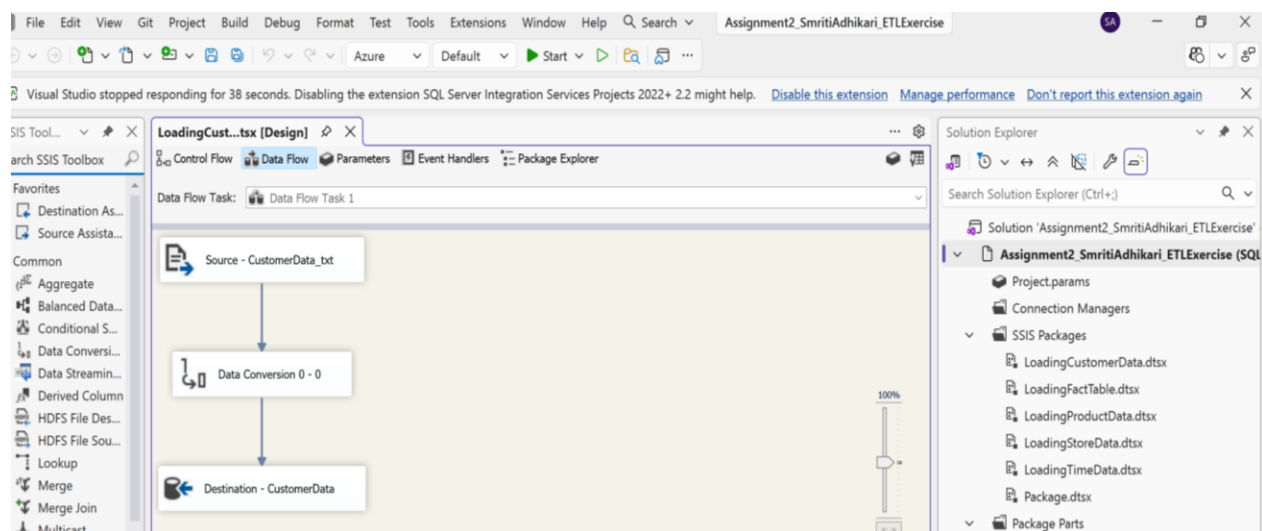
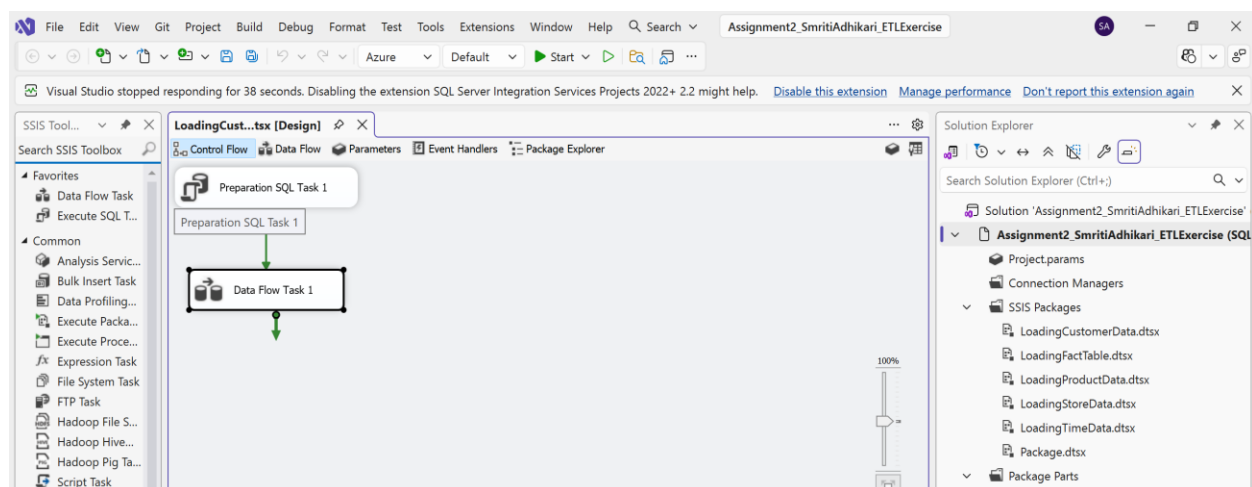
In this task, SQL Server Integration Services (SSIS) was used to perform the Extract, Transform, and Load (ETL) process. Raw data was extracted from flat files (text and CSV formats), transformed as required, and loaded into the SQL Server database Assignment2DB. Separate SSIS packages were created in Visual Studio 2022 for each dataset using the SSIS Import and Export Wizard and necessary data type conversion were made.

Input File	Package Name	Destination Table
CustomerData.txt	LoadingCustomerData.dtsx	DimCustomer
ProductData.csv	LoadingProductData.dtsx	DimProduct
StoreData.csv	LoadingStoreData.dtsx	DimStore
TimeData.txt	LoadingTimeData.dtsx	DimTime
FactTable.txt	LoadingFactData.dtsx	FactTable

Destination Table	Column	Data Type
DimCustomer	CustomerKey	Integer
DimCustomer	DateOfBirth	Date
DimProduct	ProductKey	Integer
DimStore	StoreKey	Integer
DimTime	DateKey	Integer

DimTime	FullDateAlternateKey	Date
DimTime	Day / Month / Quarter / Year	Integer
FactTable	CustomerKey, ProductKey, StoreKey, DateKey	Integer
FactTable	SalesAmount	Float
FactTable	Quantity	Integer

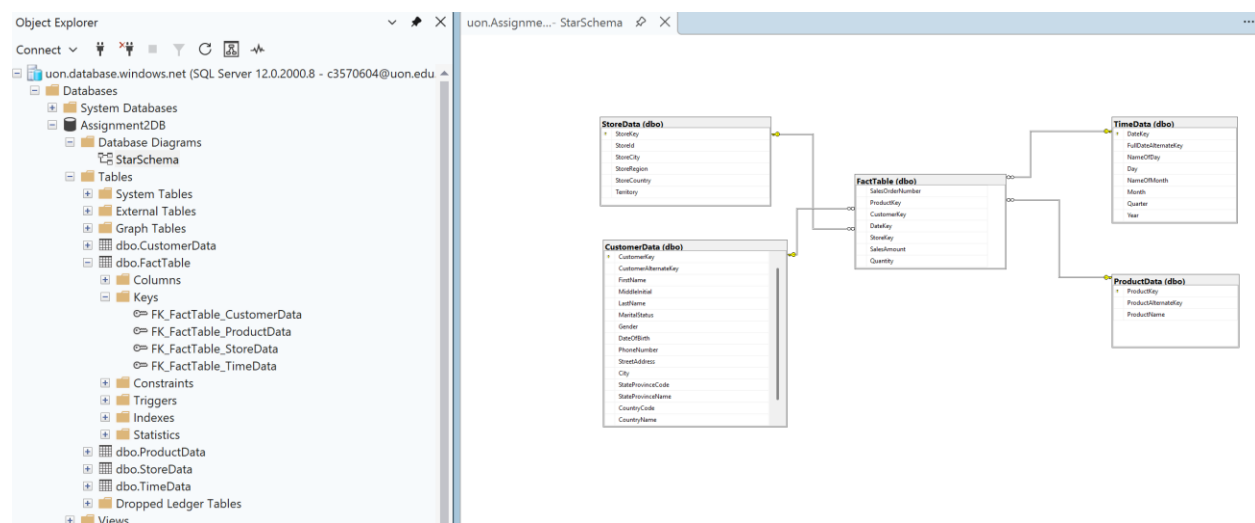
All packages were executed successfully, and both Control Flow and Data Flow components were confirmed. Finally, the loaded data was validated in SQL Server to ensure accuracy and completeness, preparing the database for further stages such as star schema design and cube modeling.



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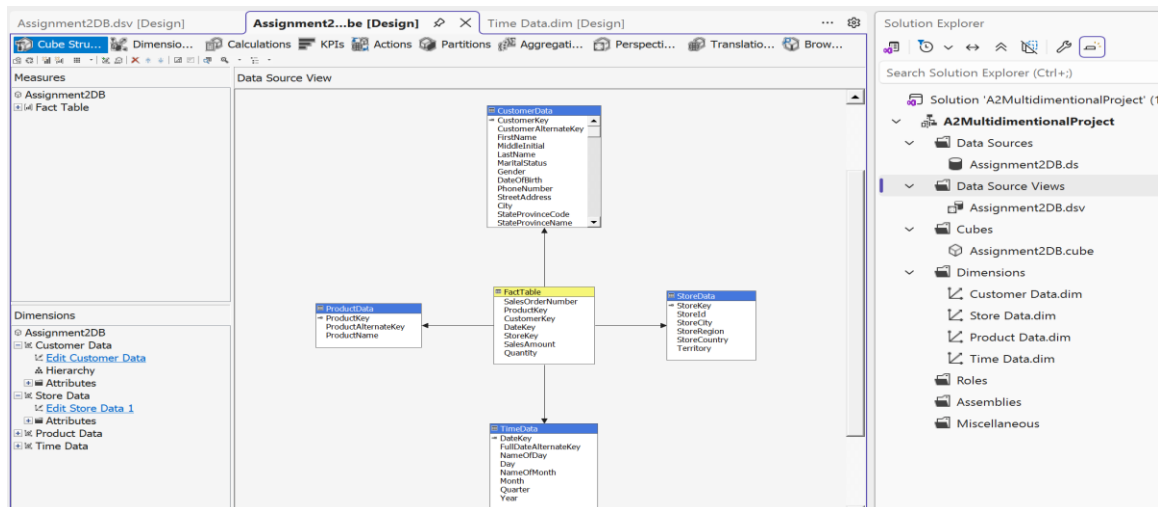
Star Schema Design

After loading the data into SQL Server, the tables were initially not related. To establish relationships and support efficient querying, a star schema was implemented using SQL Server Management Studio. The schema consists of a central fact table (FactTable) connected to four dimension tables: DimCustomer, DimProduct, DimStore, and DimTime. Primary keys were defined in each dimension table, and corresponding foreign keys were established in the fact table to show proper relationships between the tables. A database diagram was created to visually represent the schema and was exported as a PDF file for submission.



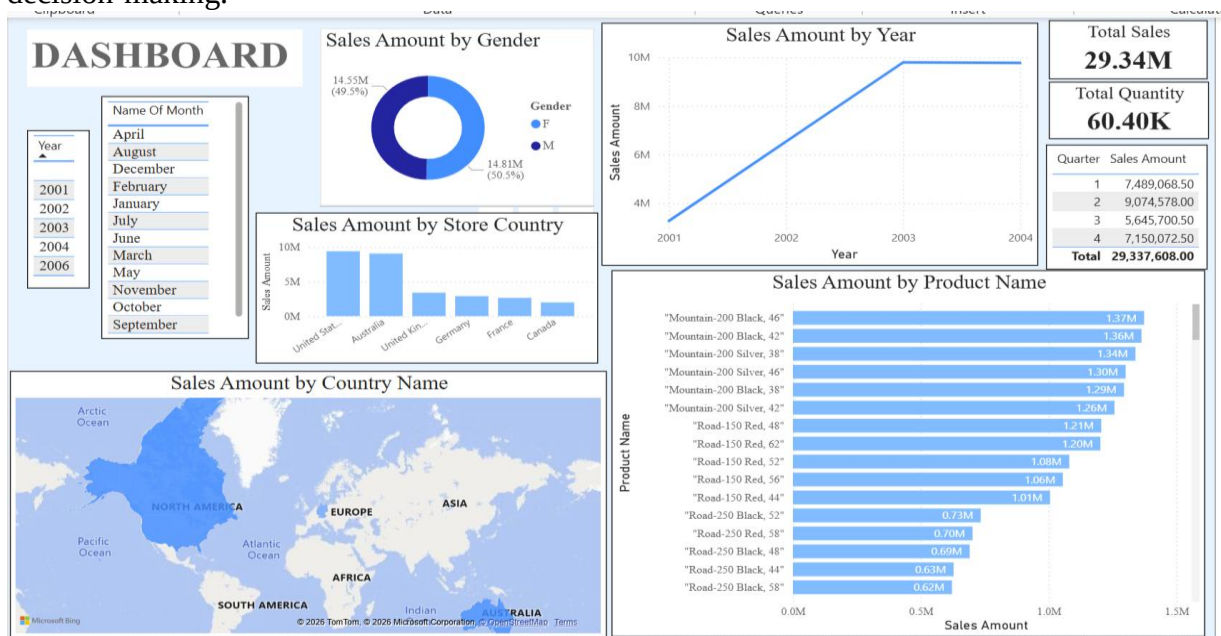
Cube Modeling

A multidimensional cube was created using SQL Server Analysis Services (SSAS) in Visual Studio 2022 under the project named A2MultidimensionalProject. First, a data source was established by connecting to the Assignment2DB database. After that, a Data Source View was created, which included all the relevant tables required for analysis. Based on the view, dimensions were defined for Customer, Product, Store, and Time. A cube model was then created by selecting appropriate measures, including SalesAmount and Quantity. Finally, the project was successfully built and deployed to the local SSAS multidimensional server instance. This made the cube available for use in Power BI, enabling further data visualization and reporting.



Power BI Dashboard

A Power BI dashboard was created using the SSAS cube as the data source. The dashboard includes multiple visuals to analyze business performance. These visuals help identify trends, patterns, and key performance indicators. The dashboard provides meaningful insights for decision-making.



Summary

This report describes the development of a business intelligence pipeline using SSIS, SSAS, and Power BI. Data was extracted from flat files, transformed, and loaded into a SQL Server database. A star schema was created to organize the data, and a multidimensional cube was built for analysis. Finally, a Power BI dashboard was developed to visualize the data and provide useful insights for decision-making.